Model setup

Geometry, material properties and loading

Atomistic parameterization

Mechanical equilibrium

Plane stress; strain field

Local transport parameters

$E_f,\ E_m,\ D$

Coupled vacancy transport and electrostatics

Convergence verification

Mesh and time step

Results and analysis

Evolution, steady states and parameter effects

Atomistic parameterization

Formation energy

Strain-gradient calculations

$E_f^0,\ A_f$

Migration barrier

CI-NEB calculations

$E_m^0,\ A_m$

Vacancy diffusion

MD
Arrhenius analysis

D_0

Parameter transfer

Atomistic parameters + computed strain field

$$E_f = E_f^0 + A_f\varepsilon_{yy} \qquad E_m = E_m^0 + A_m\varepsilon_{yy} \qquad D = D_0\exp[-E_m/(RT)]$$

Coupled transport and electrostatics

Initial and boundary conditions

Uniform, charge-neutral state

Blocking and insulating boundaries

Vacancy transport

$$J = -D\left[\frac{\partial c}{\partial x} + \frac{c}{RT}\frac{\partial E_f}{\partial x} + \frac{z_V F c}{RT}\frac{\partial \phi}{\partial x}\right]$$

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x}$$

Space charge

Electrostatic drift

Electrostatics

$$-\nabla\cdot(\epsilon\nabla\phi) = \rho$$

$$\rho = F(z_V c - c_b)$$

Fully coupled time-dependent solution